

3.1.4 Research methods

Content	Additional information
Formulation of testable hypotheses	Null hypothesis and alternative hypothesis.
Types of variable	Independent variable, dependent variable, extraneous variables.
Sampling methods	Target populations, samples and sampling methods and how to select samples using these methods: • random • opportunity • systematic • stratified. Strengths and weaknesses of each sampling method. Understanding principles of sampling as applied to scientific data.

Content	Additional information
Designing research	Quantitative and qualitative methods: - the experimental method (experimental designs, independent groups, repeated measures, matched pairs, including strengths and weaknesses of each experimental design) - laboratory experiments - field and natural experiments - interviews - questionnaires - case studies - observation studies (including categories of behaviour and interobserver reliability). Strengths and weaknesses of each research method and types of research for which they are suitable.
Correlation	An understanding of association between two variables and the use of scatter diagrams to show possible correlational relationships. The strengths and weaknesses of correlations. Computation of formulae is not required.
Research procedures	The use of standardised procedures, instructions to participants, randomisation, allocation to conditions, counterbalancing and extraneous variables (including explaining the effect of extraneous variables and how to control for them).
Planning and conducting research	How research should be planned, taking into consideration the reliability and/or validity of: - sampling methods - experimental designs - quantitative and qualitative methods.
Ethical considerations	Students should demonstrate knowledge and understanding of: - ethical issues in psychological research as outlined in the British Psychological Society guidelines - ways of dealing with each of these issues.

3.1.4.1 Data handling

Content	Additional information
Quantitative and qualitative data	The difference between quantitative and qualitative data.
Primary and secondary data	The difference between primary and secondary data.
Computation	Recognise and use expressions in decimal and standard form: use ratios, fractions and percentages, estimate results, find arithmetic means and use an appropriate number of significant figures.
Descriptive statistics	Understand and calculate mean, median, mode and range.
Interpretation and display of quantitative data	Construct and interpret frequency tables and diagrams, bar charts, histograms and scatter diagrams for correlation.
Normal distributions	The characteristics of normal distribution.